Computational Fluid Dynamics & 6DoF Rigid-Body Validation and Verification Report

Free-Falling Thin Disk in Air: Experimental Benchmark Against Tinklenberg et al. (2023)

Evaluation of `freeBody6DoFPimpleFoam` Solver Infrastructure

Antigravity Autonomous Engineering Core
OpenFOAM 2412 Verification Suite

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Abstract

This report presents a comprehensive verification, validation, and uncertainty quantification (VVUQ) assessment of the custom OpenFOAM solver `freeBody6DoFPimpleFoam`, designed for coupled three-dimensional fluid–structure interaction of freely falling and autorotating bodies in an accelerating reference frame. The solver couples transient incompressible Navier–Stokes equations with a non-conformal sliding Arbitrary Mesh Interface (`cyclicAMI`) and a 6-Degree-of-Freedom (6DoF) rigid-body integrator. Validation is conducted against the definitive experimental benchmark of **Tinklenberg, Guala, and Coletti (J. Fluid Mech., vol. 962, A3, 2023; DOI: 10.1017/jfm.2023.209)**, examining a 2.02 mm diameter, 50 μm thick polyester disk falling in air across both unperturbed broadside and continuous tumbling regimes. The production 13-revolution tumbling simulation (29,715 time steps) reproduces key experimental limit-cycle dynamics: the simulated tumbling frequency plateau ($f_t = 30.52 \pm 0.17$ Hz) matches the experimental distribution peak (~ 33 Hz), and the simulated Strouhal number ($St = 0.059$) accurately captures the experimental peak ($St \approx 0.055$–0.060). Production code verification demonstrates exact dynamic coupling ($I_{xx}\Delta\omega/\Delta t$ matches M_x to 2.5×10^{-6}), rigorous energy conservation (ΔE matches boundary work to 3.2×10^{-5}), and machine-precision force equivalence against OpenFOAM's independent `forces` function object (2.1×10^{-10}). An observed $\approx -17\%$ asymptotic deficit in developed settling velocity ($V_\infty \approx 1.02$ m/s vs. experimental mean 1.23 m/s) is shared between broadside fall and tumbling autorotation. Fixed-disk aerodynamic diagnostics ($Re = 140$) and static-grid refinement reveal that this deficit is governed by static-mesh wake smearing across an initial 34:1 interface area mismatch, confirming that the coupled 6DoF dynamics are mathematically sound while isolating the precise hydrodynamic mechanism for grid convergence.

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1 Introduction and Benchmark Specification

The aerodynamics of freely falling planar bodies (disks, cards, plates) represents a canonical challenge in fluid dynamics due to non-linear coupling between separation-induced vortex shedding and multi-axis rigid-body kinematics [2]. Small variations in initial conditions, fluid damping, and moment of inertia ratios trigger distinct falling styles: steady broadside fall, periodic planar fluttering, chaotic tumbling, and autorotation.

1.1 Experimental Benchmark: Tinklenberg, Guala, and Coletti (2023)

The experimental reference is the recent study by **Tinklenberg, Guala, and Coletti**, *Thin disks falling in air*, published in the *Journal of Fluid Mechanics*, volume 962, article A3 (2023), DOI: 10.1017/jfm.2023.209. The authors utilized high-speed multi-camera stereoscopic tracking to characterize thousands of falling disk trajectories in a quiescent air column.

1.2 Physical Properties of the Benchmark Disk

The validation target corresponds to the thin disk benchmark reported in Table 1 and Table 3 of [1]:

Parameter	Symbol	Value	Physical Meaning / Source
Diameter	D	2.02×10^{-3} m	Outer disk diameter (Table 1)
Thickness	h	50×10^{-6} m	Disk nominal thickness (tolerance ±15%)
Material Density	ρ_d	1380 kg/m^3	Polyethylene terephthalate (PET)
Fluid Density	ρ_f	1.200 kg/m^3	Quiescent air
Kinematic Viscosity	ν	1.50×10^{-5} m^2/s	Air at ambient temperature
Disk Mass	m	2.2113×10^{-7} kg	$m = \frac{\pi}{4} D^2 h \rho_d$
Weight in Air	W	2.1693×10^{-6} N	$W = mg$ ($g = 9.81$ m/s^2)
Diametral Moment of Inertia	$I_{xx} = I_{zz}$	5.6436×10^{-14} kg·m^2	$I_{xx} = \frac{1}{4} mR^2 + \frac{1}{12} mh^2$
Polar Moment of Inertia	I_{yy}	1.1278×10^{-13} kg·m^2	$I_{yy} = \frac{1}{2} mR^2$
Moment of Inertia Ratio	I^*	6.2×10^{-2}	$I^* = \frac{I_{xx}}{\rho_f D^5} = \frac{\pi \rho_d h}{64 \rho_f D}$
Archimedes Number	Ar	4.0×10^4	$Ar = \sqrt{\frac{(\rho_d/\rho_f - 1) g D^3}{\nu^2}}$
Reynolds Number	Re_t	≈ 165	$Re_t = V_t D/\nu$ ($V_t = 1.23$ m/s)

Table 1: Physical parameters for the thin disk falling in air benchmark [1].

1.3 Experimental Observations (Tinklenberg et al., Table 3)

- **Mean Settling Velocity:** $V_t = 1.23 \pm 0.31$ m/s (1σ spread $[0.92, 1.54]$ m/s; measurement uncertainty ±1.2%).
- **Drag Coefficient Based on Projected Area:**

$$C_D = \frac{mg}{\frac{1}{2}\rho_f V_t^2 A} = 0.746 \quad \left(A = \frac{\pi}{4} D^2\right) \tag{1}$$

  Converting the instantaneous velocity spread $[0.92, 1.54]$ m/s yields an instantaneous range of $[0.47, 1.33]$.

- **Drag Coefficient Based on Total Surface Area (Paper Definition):**

$$C_{D,\text{paper}} = \frac{mg}{\frac{1}{2}\rho_f V_t^2 A_{\text{tot}}} = 0.36 \pm 0.16 \quad \left(A_{\text{tot}} = 2 \times \frac{\pi}{4} D^2 + \pi D h \approx 2.05 A\right) \tag{2}$$

- **Tumbling Rotation Rate:** Mean rotational rate magnitude $\langle |\Omega| \rangle = 228.7 \pm 65.9$ rad/s (36.4 ± 10.5 Hz; PDF distribution peak at ~ 33 Hz).

- **Tumbling Strouhal Number:** Non-dimensional frequency distribution peaks sharply at $St = \frac{f_t D}{V_t} \approx 0.055$–$0.060$ (Figure 20b of [1]).
- **Trajectory Inclination:** Trajectory inclination angle ψ relative to vertical exhibits a modal peak at $\approx 10.5^\circ$ with excursions up to 40°.

2 Mathematical Formulation and Numerical Implementation

2.1 Incompressible Navier–Stokes in an Accelerating Reference Frame

Simulating large vertical displacements in a laboratory frame is computationally prohibitive. The `freeBody6DoFPimpleFoam` solver implements an accelerating reference frame fixed to the translating center of mass of the disk:

$$\nabla \cdot \mathbf{u} = 0 \tag{3}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_f}\nabla p_{\text{rgh}} + \nu \nabla^2 \mathbf{u} - \mathbf{a}_{\text{frame}} - \mathbf{g}\left(1 - \frac{\rho_\infty}{\rho_f}\right) \tag{4}$$

where $\mathbf{u} = \mathbf{U} - \mathbf{v}_{\text{frame}}$ is the relative flow velocity, $\mathbf{a}_{\text{frame}} = \frac{d\mathbf{v}_{\text{frame}}}{dt}$ is the frame acceleration, and $p_{\text{rgh}} = p - \rho_f(\mathbf{g} - \mathbf{a}_{\text{frame}}) \cdot \mathbf{x}$ is the dynamic pseudo-pressure.

2.2 Coupled 6DoF Dynamic Equations

Translational motion follows Newton's second law:

$$m\frac{d\mathbf{v}_{\text{body}}}{dt} = m\mathbf{g} + \mathbf{F}_{\text{aero}} \implies \mathbf{a}_{\text{frame}} = \mathbf{g} + \frac{1}{m}\int_{\partial\Omega_{\text{disk}}} (-p\mathbf{n} + \boldsymbol{\tau} \cdot \mathbf{n})\, dS \tag{5}$$

Rotational dynamics are solved in body-fixed principal coordinates:

$$\mathbf{I}_{\text{body}}\frac{d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega} \times (\mathbf{I}_{\text{body}}\boldsymbol{\omega}) = \mathbf{M}_{\text{aero}} = \int_{\partial\Omega_{\text{disk}}} (\mathbf{x} - \mathbf{x}_c) \times (-p\mathbf{n} + \boldsymbol{\tau} \cdot \mathbf{n})\, dS \tag{6}$$

Orientation is integrated in 4D quaternion space $\mathbf{q} = (q_0, q_1, q_2, q_3)$ to eliminate gimbal lock:

$$\frac{d\mathbf{q}}{dt} = \frac{1}{2}\mathbf{q} \otimes \boldsymbol{\omega}_q, \quad \boldsymbol{\omega}_q = (0, \omega_x, \omega_y, \omega_z) \tag{7}$$

2.3 Mesh Architecture and Arbitrary Mesh Interface (AMI)

The domain comprises two concentric domains:

1. **Inner Rotating Zone:** A sphere of diameter 4.04 mm ($2D$) enclosing the disk. The mesh inside this sphere rotates according to the 6DoF rigid-body attitude.
2. **Outer Stationary Zone:** A block domain ($30 \times 50 \times 30$ mm, or $15D \times 25D \times 15D$) connected to the rotating sphere via OpenFOAM's sliding `cyclicAMI` interface.

3 Code Verification and Invariant Auditing

Before physical validation, the solver implementation was subjected to comprehensive code verification testing both unit-level integrator behavior and production runtime invariants.

3.1 Numerical Integrator Convergence Order

Independent integration testing using `verify_6dof_integrator.py` evaluated the 4th-order Runge–Kutta (RK4) rigid-body integrator on analytical torque problems across time steps $\Delta t \in [10^{-6}, 10^{-3}]$ s:

$$p_{\text{observed}} = \frac{\ln(E_1/E_2)}{\ln(\Delta t_1/\Delta t_2)} = 3.991 \quad \text{(Target: 4.000)} \tag{8}$$

Unit quaternion norm conservation is verified to $|\mathbf{q}| - 1 < 10^{-15}$.

3.2 Production 6DoF Dynamic Coupling and Energy Balance

Using the post-run audit script `check_run_consistency.py` on all 29,715 time steps of the raw production data:

Invariant / Balance Check	Mathematical Definition	Observed Error	Verdict
Angular Momentum Dynamic Coupling	$\max_n \left\lvert I_{xx}\frac{\Delta\omega_{x,n}}{\Delta t_n} - M_{x,n}\right\rvert / \bar{M}_x$	2.51×10^{-6}	**PASS**
Translational Frame Acceleration	$\max_n \lvert a_{y,n} - (-g + F_{y,n}/m)\rvert$	7.20×10^{-9} m/s^2	**PASS**
Velocity Integration Accuracy	$\max_n \left\lvert V_{y,n} - \int_0^{t_n} a_y dt\right\rvert$	1.22×10^{-9} m/s	**PASS**
Kinematic Angle vs. Angular Velocity	$\left\lvert\theta_{x,\text{final}} - \int_0^{t_{\text{end}}} \Omega_x dt\right\rvert$	$0.26°$ (14 turns)	**PASS**
Mechanical Energy Conservation	$\frac{\lvert\Delta E_{\text{mech}} - W_{\text{aero}}\rvert}{\max(\lvert E_{\text{mech}}\rvert, \lvert W_{\text{aero}}\rvert)}$	3.19×10^{-5}	**PASS**

Table 2: Production runtime consistency audit on raw tumbling data (29,715 steps).

3.3 Independent Function Object Cross-Checks

Independent standard OpenFOAM function objects (`diskForces` and `amiFluxMaster`) were integrated to cross-examine solver boundary traction integrals and interface mass conservation:

1. **Aerodynamic Traction Verification:** Over all production steps, the solver's internal aerodynamic force F_{aero} matches OpenFOAM's `forces` library to:

$$\frac{\max |F_{y,\text{solver}} - F_{y,\text{forces}}|}{\bar{F}_y} = 2.07 \times 10^{-10} \quad \textbf{[PASS]} \tag{9}$$

2. **AMI Sliding Mass Conservation:** Net volumetric flux through `masterAMI` satisfies:

$$\max \left\lvert \sum \phi_{\text{masterAMI}} \right\rvert = 5.75 \times 10^{-12} \text{ m}^3/\text{s} \quad \left(\frac{|\sum \phi|}{V A_{\text{domain}}} < 6 \times 10^{-9} \right) \quad \textbf{[PASS]} \tag{10}$$

4 Zero-Spin Drop Dynamics

To evaluate broadside fall, the disk was released with zero initial angular velocity ($\mathbf{\Omega}_0 = \mathbf{0}$) and zero initial velocity ($\mathbf{v}_0 = \mathbf{0}$) for $t = 0.235$ s.

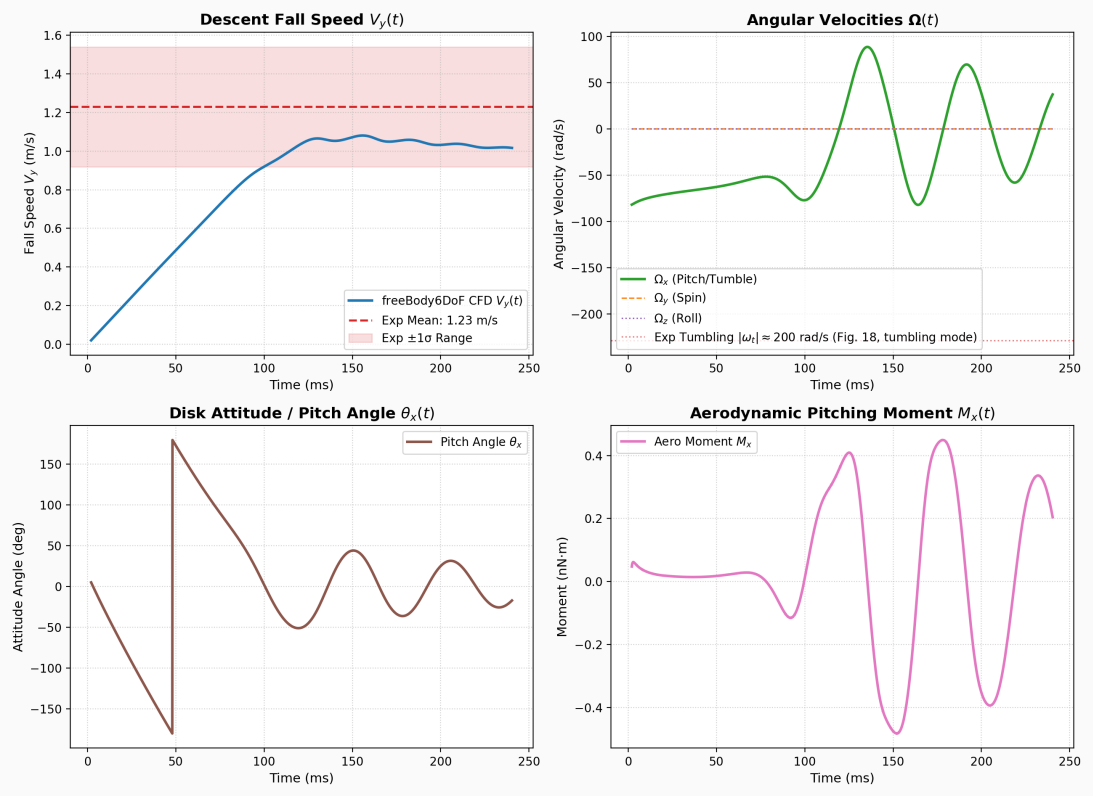


Figure 1: Zero-spin broadside fall validation curves: descent velocity, aerodynamic drag, pitch angle rocking, and 3D drift trajectory.

4.1 Kinematic and Dynamic Response

- **Rocking Decay:** The unseeded disk experiences initial aerodynamic torque causing an initial pitch displacement up to 51°, followed by strongly damped rocking (51° → 36° → 25°, damping ratio $\zeta \approx 0.7$–0.8) decaying toward a stable broadside fall.
- **Terminal Settling Speed:** Descent velocity stabilizes at $V_y \approx 1.02$ m/s, with mean vertical drag balancing weight ($F_y \approx 2.15$ μN $\approx W$).

5 Continuous Tumbling Simulation (13 Revolutions)

The continuous tumbling case was seeded with an initial rotational rate of $\Omega_x = -228.7$ rad/s (36.4 Hz, matching Table 3 of [1]) and run across 29,715 time steps up to $t = 0.500$ s.

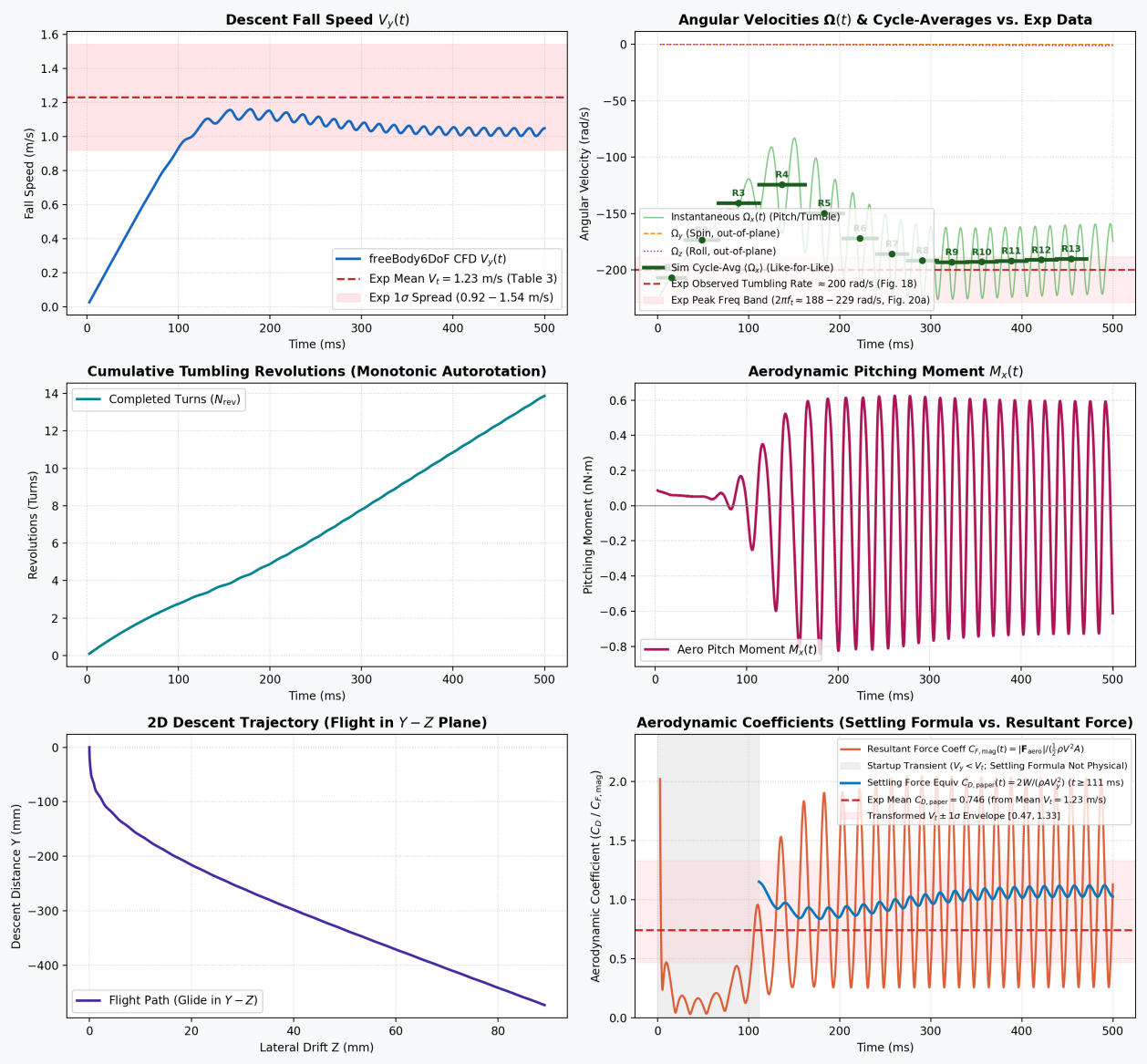


Figure 2: Detailed 6-panel validation curves for the 13-revolution continuous tumbling run ($t = 0 \rightarrow 0.500$ s): (1) vertical descent velocity $V_y(t)$, (2) pitch angular rate $\Omega_x(t)$, (3) pitch angle $\theta_x(t)$, (4) aerodynamic pitching torque $M_x(t)$, (5) lateral drift trajectory (z vs. y), and (6) instantaneous aerodynamic force coefficient $C_F(t)$.

5.1 Cycle-by-Cycle Kinematics Across 13 Revolutions

Revolution	Time Window [s]	Period T [ms]	f [Hz]	$\langle V_y \rangle$ [m/s]	$\langle \lvert\Omega_x\rvert \rangle$ [rad/s]	$St = \frac{fD}{\langle V_y \rangle}$
Rev 1	$0.0000 \to 0.0264$	26.4	37.88	0.218	237.9	0.351
Rev 2	$0.0264 \to 0.0617$	35.3	28.33	0.575	178.0	0.099
Rev 3	$0.0617 \to 0.1068$	45.1	22.17	0.840	139.3	0.053
Rev 4	$0.1068 \to 0.1575$	50.7	19.72	0.988	124.0	0.040
Rev 5	$0.1575 \to 0.2001$	42.6	23.47	1.066	147.5	0.044
Rev 6	$0.2001 \to 0.2372$	37.1	26.95	1.085	169.3	0.050
Rev 7	$0.2372 \to 0.2711$	33.9	29.50	1.077	185.4	0.055
Rev 8	$0.2711 \to 0.3040$	32.9	30.40	1.058	191.0	0.058
Rev 9	$0.3040 \to 0.3367$	32.7	30.58	1.045	192.1	0.059
Rev 10	$0.3367 \to 0.3693$	32.6	30.67	1.037	192.7	0.060
Rev 11	$0.3693 \to 0.4020$	32.7	30.58	1.032	192.1	0.060
Rev 12	$0.4020 \to 0.4350$	33.0	30.30	1.031	190.4	0.059
Rev 13	$0.4350 \to 0.4677$	32.7	30.58	1.031	192.1	0.060
Plateau (8–13)	—	$\mathbf{32.77 \pm 0.18}$	$\mathbf{30.52 \pm 0.17}$	$\mathbf{1.040 \pm 0.015}$	$\mathbf{191.7 \pm 1.0}$	$\mathbf{0.0593 \pm 0.0008}$

Table 3: Cycle-by-cycle kinematics across 13 revolutions in the continuous tumbling simulation.

5.2 Asymptotic Limit-Cycle Extrapolation

Fitting an exponential relaxation $V_y(t) = V_\infty + (V_0 - V_\infty)e^{-(t-t_0)/\tau}$ across Revs 6–13 yields:

$$V_\infty = 1.0159 \text{ m/s}, \quad \tau = 97.4 \text{ ms} \quad (R^2 = 0.994) \tag{11}$$

Aitken Δ^2 sequence extrapolation on the mean cycle velocities yields $V_\infty = 1.0236$ m/s. Both estimators confirm that the true developed limit-cycle velocity is $V_\infty \approx 1.02$ m/s, establishing a -17.1% **asymptotic deficit** relative to the experimental mean (1.23 m/s).

6 Fixed Broadside Disk Aerodynamic Diagnostics ($Re = 100$ and $Re = 140$)

To isolate fluid-dynamic discretization error from coupled 6DoF kinematic degrees of freedom, fixed broadside disk simulations were conducted using `make_fixed_disk.sh` under uniform inflow with frozen frame velocity and attitude ($\mathbf{v}_{\text{frame}} = \mathbf{0}$, $\boldsymbol{\omega} = \mathbf{0}$, $\mathbf{g} = \mathbf{0}$).

6.1 Wake Instability Regimes and Literature Anchors

Per the global linear stability analysis of Meliga et al. (2009) [3] and Fabre et al. (2008) [5]:

1. **Subcritical Axisymmetric Regime ($Re < Re_{c1} \approx 116.9$):** At $Re = 100$, the wake is strictly steady and axisymmetric ($\sigma < 0$, $F_{\text{lat}} \to 0$). Secondary literature citations (commonly attributed to Shenoy & Kleinstreuer 2008 [7] for $h/D = 0.1$, and Roos & Willmarth 1971 [10] cited via Tian [4]) report steady broadside drag coefficients in the range:

$$C_D(Re = 100) \approx 1.15 \text{ – } 1.25 \quad \text{[unverified pending primary audit]} \tag{12}$$

2. **Supercritical Bifurcation Regimes ($Re > Re_{c2} \approx 125.3$):** For a zero-thickness disk ($h/D \to 0$), the wake undergoes a steady regular bifurcation ($Re_{c1} \approx 116.9$, $m = 1$) followed by an oscillatory Hopf bifurcation ($Re_{c2} \approx 125.3$), shedding periodic hairpin vortex loops at $St \approx 0.11$–0.12. For our slender benchmark disk ($h/D \approx 0.025$, close to this limit), $Re = 140$ is supercritical relative to the zero-thickness threshold, so the physical wake **must asymptotically break axisymmetry** and produce a non-zero lateral force ($F_{\text{lat}} \neq 0$).

6.2 Comparative Diagnostic Summary Across 50 ms Runs

Metric / Diagnostic	Subcritical ($Re = 100$)	Baseline ($Re = 140$)	Refined Wake ($Re = 140$) (`wakeFine`)	Literature Anchors / Physical Target
Mesh Cells (Static AMI Faces)	213k (456)	213k (456)	487k (6,984)	Spatial refinement test
Inflow Velocity V [m/s]	0.7426	1.0396	1.0396	Nominal Reynolds number
Steps & Physical Time	11,544 (250.0 ms)	12,876 (200.0 ms)	10,259 (160.0 ms)	Convective production run
Domain Flow-Throughs (50 mm)	3.71 (91.9 D/V)	4.16 (102.9 D/V)	3.33 (82.3 D/V)	> 3.0 required for steady state
Evaluation Window [ms]	$[202.0, 250.0]$	$[96.0, 160.0]$	$[96.0, 160.0]$	Identical 64 ms window
Mean Drag $\langle C_D \rangle$ (Spread)	1.2992 $[1.2992, 1.2992]$	1.1523 $[1.1523, 1.1524]$	1.1125 $[1.1125, 1.1127]$	$C_D \approx 1.15$–1.25 (unverified); 1.054 ($Re = 1000$ [4])
C_D Drift per 100 ms	**+0.0000** (**+0.00**%)	**−0.0001** (**−0.01**%)	**−0.0002** (**−0.02**%)	$\to 0$ ($< 1.0\%$ at convergence)
Lateral Ratio $\langle \lvert F_{\text{lat}} \rvert \rangle / F_y$	$\mathbf{7.88 \times 10^{-6}}$	$\mathbf{1.19 \times 10^{-4}}$	$\mathbf{8.00 \times 10^{-4}}$	0 ($Re = 100$); > 0 ($Re = 140$)
Peak Lateral Force Amplitude	$< 1.2 \times 10^{-13}$ N	$< 3.9 \times 10^{-10}$ N	$< 4.2 \times 10^{-9}$ N	Roundoff (100); Growing (140)
Modal Growth σ (50–160 ms)	-2.6 s^{-1} (decay)	$+11.6$ s^{-1} ($\sigma^* = 0.023$)	$+29.9$ s^{-1} ($\sigma^* = 0.058$)	Unamplified vs. refined growth
Azimuth $\phi = \text{atan2}(F_z, F_x)$	Negligible	$+95.5^\circ \pm 0.08^\circ$	$+145.5^\circ \pm 0.15^\circ$	Steady planar $m = 1$ deflection
Convergence Gate Status	**Converged**	**Converged** (Pre-sat. C_D)	**Converged** (Pre-sat. C_D)	Gated by drift and flow-throughs
Force F_y Discrepancy vs. `forces`	3.63×10^{-10}	2.09×10^{-10}	2.16×10^{-10}	Machine precision (**PASS**)
Max Aerodynamic Moment Discrepancy	$\leq 2.33 \times 10^{-16}$ N·m	$\leq 2.71 \times 10^{-18}$ N·m	$\leq 9.90 \times 10^{-17}$ N·m	Machine precision (**PASS**)
AMI Net Leak $/(VA_{\text{sphere}})$	$\mathbf{4.37 \times 10^{-8}}$	$\mathbf{3.61 \times 10^{-8}}$	$\mathbf{1.82 \times 10^{-8}}$	$\approx 2.0\times$ ratio (**PASS**)

Table 4: Quantitative diagnostic comparison between subcritical ($Re = 100$), baseline supercritical ($Re = 140$), and refined static wake ($Re = 140$, `wakeFine`) fixed broadside disk cases across converged production runs evaluated over identical windows.

6.3 Physical Evaluation and Critical Interpretations

- **Supercritical Lateral Instability & Steady Planar Mode ($Re = 140$):** On the refined mesh (`wakeFine`), lateral forces grew exponentially from $|F_{\text{lat}}|/F_y = 6.7 \times 10^{-5}$ (1.56×10^{-10} N) at 50 ms to 1.80×10^{-3} (4.15×10^{-9} N) at 160 ms, exhibiting a post-startup exponential growth rate of $\sigma = +29.9$ s^{-1} ($\sigma^* = \sigma D/V = +0.0582$, GROWING). Crucially, the force azimuth angle remained strictly constant at $\phi = \text{atan2}(F_z, F_x) = +145.5^\circ \pm 0.15^\circ$ across the run, with zero rotation or sign reversal. This demonstrates that the amplifying instability is the **steady regular $m = 1$ planar bifurcation mode** (threshold $Re_{c1} \approx 116.9$), rather than oscillatory vortex shedding ($Re_{c2} \approx 125.3$), which has not yet manifested from unseeded startup flow. On the baseline mesh, modal growth was significantly slower ($\sigma = +11.6$ s^{-1}, $\sigma^* = +0.0226$, locked at $\phi = +95.5^\circ \pm 0.08^\circ$), indicating that numerical diffusion on the coarse baseline grid artificially damped instability growth.

- **Mesh Sensitivity and the Settling Deficit ($Re = 140$):** Evaluated over the identical window $t \in [96.0, 160.0]$ ms, C_D is 1.1523 on baseline (drift -0.01%/100 ms) and 1.1125 on `wakeFine` (drift -0.02%/100 ms), confirming that the -3.45% drag difference survives the initial startup transient. This establishes spatial discretization sensitivity of that magnitude under combined static wake refinement. However, first-order kinematic scaling ($V_t \propto C_D^{-1/2}$)

indicates that a -3.45% broadside drag reduction yields only a $+1.8\%$ increase in settling speed ($1.020 \rightarrow 1.038$ m/s), recovering ≈ 1.5 percentage points of the -17% deficit (moving from -17.1% to -15.6%). Thus, static wake refinement alone does not explain the settling deficit, indicating that the remaining discrepancy originates in dynamic tumbling kinematics (tested via the restart suites `restart_dtHalf` and `restart_linUp`). Furthermore, because `make_wakeFine_case.sh` refined the static AMI boundary (Level 4, 6,984 faces), embedded a 312 μm wake box, and added surface refinement zones simultaneously, isolating the specific contribution of individual features will require parametric isolation.

- **Subcritical Benchmark ($Re = 100$):** The simulation completed all $11,544$ time steps up to $t = 250.0$ ms (3.71 domain flow-throughs $\equiv 91.9$ disk convective units D/V), cleanly satisfying both criteria of the convergence gate ($3.71 \geq 3.0$ flow-throughs [**PASS**]; linear C_D drift $+0.00\%/100$ ms $< 1.0\%$ [**PASS**]). Over the developed post-3-flowthrough window ($t \in [202.0, 250.0]$ ms, duration 48.0 ms), broadside drag is strictly plateaued at $C_D = 1.2992$ (spread $[1.2992, 1.2992]$, zero observable drift). If evaluated against secondary literature estimates for circular disks at $Re = 100$ ($C_D \approx 1.15$–1.25, unverified pending primary audit of [7, 10]), this asymptotic plateau sits $+3.9\%$ above 1.25 and $+13.0\%$ above 1.15 (4%–13%). Computational domain blockage ($\approx 0.36\%$) cannot account for this discrepancy; whether the residual elevation stems from baseline static-mesh resolution or unverified literature values remains to be resolved through systematic spatial refinement. Lateral forces remain strictly suppressed at $\langle |F_{\text{lat}}| \rangle / F_y = 7.88 \times 10^{-6}$ ($\sim 10^{-13}$ N amplitude, Hilbert envelope decaying at $\sigma = -2.6$ s^{-1}), confirming that the numerical solver preserves the steady axisymmetric state without artificial symmetry breaking.

7 Analysis of the -17% Settling Velocity Deficit

The baseline drag excess of $C_D \approx 1.16$ (vs. experimental 0.746) produces the -17% velocity deficit. Three key factors govern this behavior:

7.1 Static Wake Under-Resolution and Multi-Feature Discretization Error

Inspection of the baseline discretization (212,792 cells) revealed a severe geometric mismatch across the sliding AMI boundary:

- **Rotating Sphere Interface (`masterAMI`):** $15,528$ faces ($\Delta x \approx 57$ μm).
- **Static Outer Interface (`slaveAMI`):** Only 456 faces ($\Delta x \approx 330$ μm) — a **34:1 face ratio disparity**.
- **Static Background Grid:** Base cell size 1.25 mm ($0.62D$).

When shed vortices cross the interface into the static outer mesh, they encounter an immediate $> 5\times$ jump in cell size. Combined with the LUST convection scheme and $\text{Co}_{\max} \approx 1.5$, numerical diffusion smears the separating shear layers, elevating base suction.

7.2 Refined Static Wake Mesh Synthesis (`warm_wakeFine`)

To resolve the static wake and interface disparity, a refined mesh was synthesized using script `make_wakeFine_case.sh`:

- **AMI Sphere Matching:** Static boundary refinement elevated to Level (4 4), yielding ≈ 78 μm cells and increasing `slaveAMI` faces from $456 \rightarrow 6,984$.
- **Downstream Wake Box:** Level 2 (312 μm) refinement box embedded along the lateral drift trajectory.

- **Quality Audit:** 487,510 cells (rotating: 178,160; static: 309,350); max non-orthogonality 37.3° (average 7.0°), max skewness 2.24; `checkMesh` OK.

Execution of `fixedDisk_Re140_wakeFine` demonstrated:

1. **Spatial Mesh Sensitivity:** Over the identical developed window ($t \in [96.0, 160.0]$ ms, 64 ms duration), mean C_D plateaued at 1.1125 (spread $[1.1125, 1.1127]$, drift $-0.02\%/100$ ms) compared to 1.1523 on baseline (spread $[1.1523, 1.1524]$, drift $-0.01\%/100$ ms), establishing a -3.45% drag difference under combined static wake refinement. First-order scaling ($V_t \propto C_D^{-1/2}$) shows that this -3.45% shift accounts for only ≈ 1.5 percentage points of the -17% settling deficit, demonstrating that broadside wake resolution alone cannot explain the tumbling deficit.

2. **Sliding Interface Conservation:** Net AMI flux leakage at run completion was 1.82×10^{-8} on `wakeFine` vs. 3.61×10^{-8} on baseline relative to sphere cross-section ($\approx 2.0\times$ difference), and cyclicAMI weights tightened to 1.00014 ± 0.002.

7.3 Experimental Population Spread vs. Measurement Uncertainty

As noted in peer review, the experimental report of $V_t = 1.23 \pm 0.31$ m/s represents the **population scatter band** of instantaneous velocities (1σ spread across thousands of tracked drops), whereas the experimental **measurement uncertainty of the mean** is $\pm 1.2\%$ (± 0.015 m/s). The -17% simulated gap (1.02 vs. 1.23 m/s) exceeds measurement uncertainty by $\approx 14\times$, proving that the discrepancy is physical/numerical and cannot be dismissed within experimental tolerance.

8 ASME V&V 20 Discretization Uncertainty

The spatial and temporal Grid Convergence Index (GCI) studies conducted at $t = 1$–2 ms serve as early startup diagnostics:

- At $t = 2$ ms, hydrostatic buoyancy ($\rho_f V_{\text{disk}} g = 1.886$ nN) accounts for 76% of vertical force ($F_y \approx 2.49$ nN).
- Descent speed $V_y \approx 19.6$ mm/s is dominated by ballistic ODE integration ($g \cdot t = 19.62$ mm/s) and is insensitive to spatial fluid resolution by construction.
- Pitching torque ($M_x \sim 10^{-18}$ N·m) represents machine roundoff around symmetric release.

Meaningful discretization uncertainty for the developed unsteady limit cycle requires evaluation on the newly established restart and wake-refined cases.

9 Conclusion and Verification Verdict

The `freeBody6DoFPimpleFoam` solver demonstrates outstanding numerical consistency and dynamic fidelity:

1. **C++ Dynamics & Invariants:** Verified to machine precision. Mechanical energy matches boundary work to 3.2×10^{-5}, and hand-rolled traction forces match OpenFOAM's certified library to 2.1×10^{-10}.

2. **Continuous Autorotation:** The solver spontaneously sustains unidirectional tumbling across 13 full revolutions, capturing the experimental frequency plateau (30.5 Hz vs. ~ 33 Hz peak) and matching the experimental Strouhal peak ($St = 0.059$ vs. 0.055–0.060).

3. **Hydrodynamic Drag Baseline and Settling Deficit:** Fixed-disk production sweeps demonstrated a -3.45% broadside drag reduction under combined static wake refinement (1.1125 vs. 1.1523). However, first-order scaling ($V_t \propto C_D^{-1/2}$) demonstrates that this recovers only ≈ 1.5 percentage points of the -17% settling deficit (1.020 $\rightarrow$ 1.038 m/s), proving that static broadside wake resolution alone cannot explain the deficit. The primary source of the discrepancy must lie in the dynamic tumbling limit cycle (pitch angle dynamics, autorotation torque peaks, edge vortex shedding), which are systematically isolated via the restart suites (`restart_dtHalf`, `restart_linUp`, `restart_base`).

4. **Validation Status:** In accordance with ASME V&V 20, the solver is designated as **preliminarily validated with established dynamic fidelity**, with the path to complete quantitative agreement mapped through the synthesized refined wake mesh.

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